

ARPIT KUMAR

Data Scientist & ML Engineer | IIT Kharagpur | NLP · LLM/RAG · MLOps · Production ML

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PROFESSIONAL SUMMARY

Fourth-year dual degree student at IIT Kharagpur (Chemical Engineering + AI) with end-to-end Machine Learning (ML) experience spanning Natural Language Processing (NLP), Computer Vision (CV), LLM/RAG engineering, and MLOps. Ranked Top 0.5% globally in Amazon ML Challenge 2025 with a multimodal price prediction system full reproducibility via SHA-256 model fingerprinting, DVC, and MLflow. Developed state-of-the-art Wav2Vec2/WavLM fusion models for Speech AI research and deployed a RAG-based LLM support platform cutting resolution time by 35%. Codeforces Expert (peak 1612), AIR 135 in Integral Cup 2026, & Gold Medalist in IIT KGP's GC Data Analytics 2025. Seeking a Data Science internship to deliver full-stack AI/ML solutions with real product impact.

WORK EXPERIENCE

AI Researcher | Internship

May 2025 - Jul 2025

Prof. Shyamal Kumar Das Mandal | Advanced Technology Development Centre | IIT Kharagpur

On-site • Kharagpur, West Bengal, IN

Native Language Identification (NLI) from L2 English Speech

- Engineered a speaker-balanced preprocessing pipeline for the NISP dataset, ensuring model fairness by standardizing **40/10/7** splits across **57** speakers per language to mitigate class imbalance and audio duration variability across **5** native Indian languages (Hindi, Telugu, Tamil, Kannada, and Malayalam).
- Developed and benchmarked seven deep learning architectures from CNN and **ResNet-18/50** baselines to advanced fusion models by extracting normalized acoustic features like **128-bin Log-Mel spectrograms**, **40-bin MFCCs** with $\Delta/\Delta-\Delta$ coefficients to capture nuanced phonetic influence in L2 English speech.
- Achieved a state-of-the-art validation accuracy of **0.88** using a **Wav2Vec2** and **WavLM-BiLSTM** fusion model with frozen transformer layers, proving that context-aware self-supervised representations provide significant stronger discriminative cues for L1 language identification than classical spectral features.
- Analyzed speaker separability and sociolinguistic variance by high-dimensional **UMAP** visualizations of embeddings, identifying overlaps between Hindi, Tamil pronunciation while quantifying tight, well-separated clusters for Telugu and Malayalam speakers for deep insights into regional language influence.

Technologies / Skills Used : CNN (ResNet), BiLSTM, Mel-Spectrograms, MFCCs, t-SNE, Wav2Vec2, WavLM

PROJECTS

Multimodal ML Platform Pipeline for E-Commerce | 📁 Github 🌐 Live Demo 🌐 Website

Sep 2025 - Dec 2025

AI Engineer | Top 0.5%, Amazon ML Challenge'25 —national competition (sponsored by Amazon)

- Built a multimodal price prediction platform achieving **25.45 SMAPE (RMSE 0.82, MAE 0.66)**, as measured on validation benchmarks, by fusing **SBERT text embeddings, CLIP image features, and engineered tabular inputs** with a **5-fold stacked ensemble (RF, XGBoost, LightGBM, CatBoost + Ridge)**.
- Eliminated training-serving skew and ensured deterministic reproducibility, as measured by consistent reload outputs across environments, by implementing **immutable run-scoped model bundles with SHA-256 fingerprinting**, packaging vectorizers, feature selectors, dimensionality reducers, and model weights.
- Architected production FastAPI serving layer featuring request-ID middleware, API-key protection, data-quality reject/quarantine policies, operational endpoints (/healthz, /readyz, /v1/predict), achieved 469.1 ms p95 end-to-end latency on 100 row smoke inference requests (2.04-39.09 ms/row observed).
- Improved experiment tracking and deployment reliability, as measured by auditable lineage across **46+ runs and automated CI/CD workflows**, by integrating **DVC, MLflow, and DagsHub** with GitHub Actions pipelines for gated promotion, registry updates, smoke testing, and production readiness validation.

Technologies / Tools Used : PyTorch, BERT, CLIP, Docker, Python, FastAPI, MLflow

LLM-Powered Employee Support Platform with RAG and Scalable AI Infrastructure | 📁 Github

Jan 2025 - Apr 2025

AI Engineer | 1st place, GC OpenSoft'25 — inter-hall software competition, IIT Kharagpur (sponsored by Deloitte)

- Developed an **LLM-powered employee support platform** that **reduced average resolution time by 35%**, as indicated by session analytics, through a RAG-based conversational pipeline utilizing **embedding stores (FAISS/Chroma)**, prompt-engineered workflows, and context-aware multi-turn interaction handling.
- Enhanced answer grounding and **decreased hallucination rate by 40%**, as assessed through human evaluation audits, by integrating vector search, retrieval pre-processing, and citation-based response generation, ensuring **evidence-backed outputs** and increased reliability across various employee support queries.
- Minimized inference latency and **boosted API throughput by 50%**, as measured by tracing metrics, through the implementation of async **FastAPI endpoints**, non-blocking backend-to-LLM communication, and optimized model routing strategies for efficient real-time response handling in production environments.
- Cut model cost per session by **30%** while enhancing observability and analytics, by applying token optimization, context summarization, and adaptive model selection, alongside containerized deployment, CI/CD pipelines, and monitoring with OpenTelemetry, Prometheus, and Grafana dashboards.

Technologies / Tools Used : FastAPI, Python, LangChain, OpenAI API, Groq, PyTorch, Docker, JWT

SKILLS

Programming Languages :

Python, C++, SQL, C, JavaScript

Data Science :

Machine Learning, Statistics, Probability, A/B Testing, Deep learning, Computer Vision, Data Analysis

Generative AI :

Retrieval-Augmented Generation, Model Interpretability, Distributed Inference, Quantization

Frameworks Libraries :

PyTorch, TensorFlow, Scikit-learn, FastAPI, Flask, JAX, Numpy, Pandas, React

Tools Platforms :

MLflow, DVC, Git, GitHub, Docker, Dagshub, CI/CD, GitHub Actions

Languages :

Hindi, English

EDUCATION

Indian Institute of Technology (IIT) Kharagpur | Kharagpur, India

Jul 2022 - Present

Integrated B.Tech - M.Tech (Dual Degree) in Chemical Engineering

GPA : 8.86/10

Micro-Specialization: Artificial Intelligence and Applications

POSITION OF RESPONSIBILITY

Developers' Society, TSG, IIT Kharagpur

Sep 2023 - Present

Advisor | Machine Learning & Software Architecture

Kharagpur, West Bengal, IN

Public Policy and Governance Society, IIT Kharagpur

May 2023 - Sep 2024

Executive Member | Quantitative Policy Research

Kharagpur, West Bengal, IN

AWARDS & ACHIEVEMENTS

- Competitive Programming:** Achieved **Expert** rating on **Codeforces** (Peak: 1612), advance proficiency in data structures and algorithmic problem-solving.
- Quant & ML Challenges:** Ranked **Top 20** national in **DTL Quant Challenge 2024** & **Top 0.5%** global **Amazon ML Challenge 2025** out of 50K+ participant.
- Mathematics:** Secured **AIR 135** in the **Integral Cup 2026** in 3 tracks Probability Theory & Statistics, Linear Algebra & Optimization and Integral Analysis.
- GATE 2026:** Achieved **AIR 807** in the **GATE 2026 CH** stream (conducted by IIT Guwahati), ranking among the top tier of national engineering graduates.
- JEE 2022:** Secured **AIR 1478** in **JEE Advanced 2022** and a **98.28 percentile** in **JEE Mains** among 1.1M+ candidates, the most competitive exam in India.
- Institute Championships:** Awarded **1 Gold Medal** (General Championship Data Analytics 2025), **2 Silver Medals** (Open IIT Data Analytics 2024 & Open IIT Case Study 2024), and **1 Bronze Medal** (General Championship ChemQuest 2025) at IIT Kharagpur.